Game Design Document

Group Members: Adam Canarick, Effie Li, Susan Xie

Game Title: Ice Gems

1 EXECUTIVE SUMMARY

Ice Gems is an action-packed, exploration game that allows anyone to enter the beautiful ice caves of the mystical Himalaya Mountains. You take on the role of a futuristic mountain-climber who's on a quest to collect all 10 magical gems located in the different ice caves found in the monstrous rock walls of the mountains. However, your ice shoes plummeted down the mountain on your ascent up to the peak, so you must use the propulsion from your jet-pack backpack to glide your way through the super slippery cave floors.

With a limited amount of resources, huge ice pillars and walls barricading your paths, freezing cold ice ponds, as well as a dangerous abominable snowman in the cave, our game asks you to strategically collect mystical gems to fuel your jetpack to journey to the back of the cave where the magical gem of that level is located.

We believe that this game uniquely challenges players' hand-eye coordination, visual perception, and understanding of physics to successfully capture all of the gems in the mountains. The simple and maze-like level structure allows for quick, fun-fueled gameplay and an ability to increase your skill-ability with each gameplay.

The game is developed purely for PC platforms. We expect end-to-end implementation to take 5 weeks or, assuming 3 part-time designers/developers 14 man-hours.

2 GAME DESIGN - CREATIVE

2.1 High Concept

Ice Gems, designed by Adam, Effie, and Susan. The player navigates through a cave full of obstacles and collects gems that could move the player forward, melt down the boss, or freeze the time. After navigating through the cave, the player wins by shooting gems at the boss to defeat it.

2.2 Design Goals

2.2.1 Main Design Features

2.2.1.1 Player goals and objectives.

Main goal is to get to the end of the cave and retrieve the special gem.
Secondary goal is to collect gems as your resources go through the cave.

Type of challenges includes obstacles. Ice poles and ice walls block your way, and ice puddles cost your gems if you accidentally slide into them. In addition, the abominable snowman roars occasionally, which moves you backward unless you hide behind a ice pole or a ice wall.

Types of conflicts include resource management. You can propel yourself forward to reach more gems but in doing so you also use gems your own.

Winning condition – you need to defeat the abominable snowman and retrieve the special gem.

2.2.1.2 Main rules and procedures

Operational rules – You can collect gems on the wall of the cave, and you could slide on the ice using your jet-pack. The more fuel you use to propel yourself, the faster you move and the more gems you use. You can shoot rubies at the abominable snowman to melt it down in order to retrieve the crystal at the end and win the game.

Main game mechanic – The character will look at the direction of the mouse, and the mouse click and hold determines how much fuel the character uses (holding the mouse click for longer indicates more fuel use). The player can throw rubies at the abominable snowman by pressing the E key. The player can collect gems by pressing F. The player can fly up using the spacebar.

Player Resources

Types of resources: Ruby, Diamond

How does the player use resources to achieve goals?
Rubies can be used to melt down ice, which is useful in defeating the abominable snowman. Every time a ruby hits the abominable snowman, the abominable snowman decreases in size. Diamonds are used to propel yourself forward.

2.2.1.3 Boundaries and Constraints

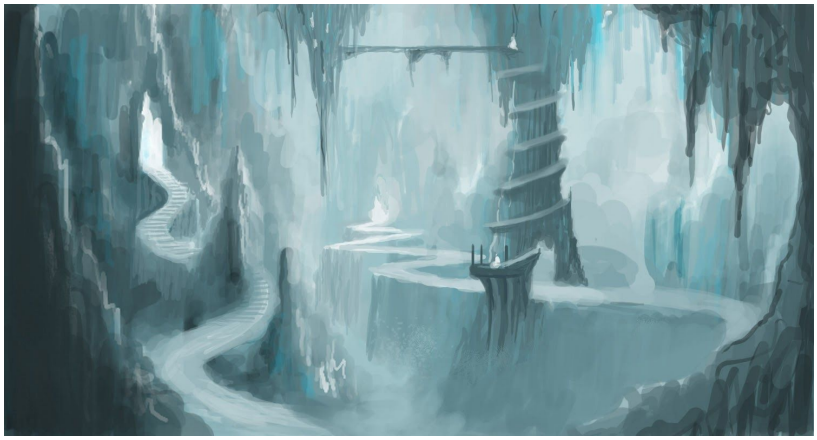
The actions are limited to the empty areas within the cave. The player cannot climb onto ice walls or ice poles. The player can only collect gems within reach, which will light up in the gameplay. The player cannot bypass the abominable snowman before melting it down.

2.2.2 Appeal

What is the game genre?
Collect and shooter
Who is the target audience?
People who are intrigued by a mission and adventure
What is their age?
Teenagers

Why is the game fun to play?
It is interesting to play because each level is different and there are many conflicts in the game. In addition, the graphics are aesthetically appealing.

2.2.3 Look and Feel

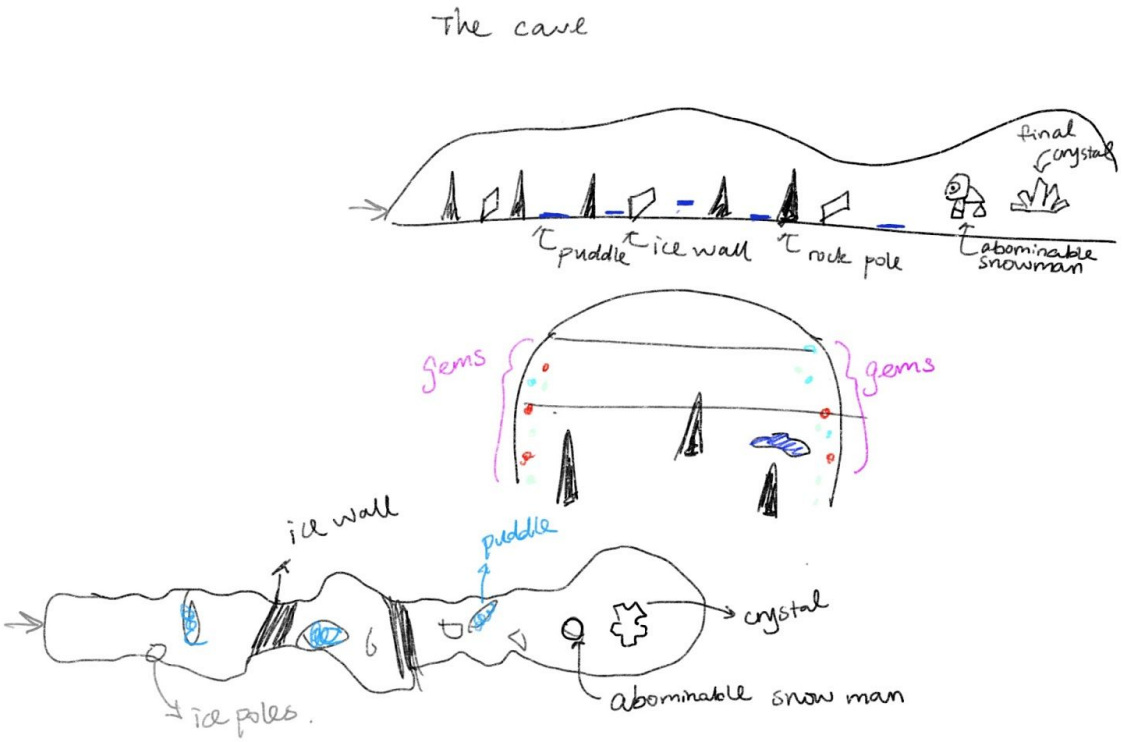


2.3 Worlds, Characters and Story

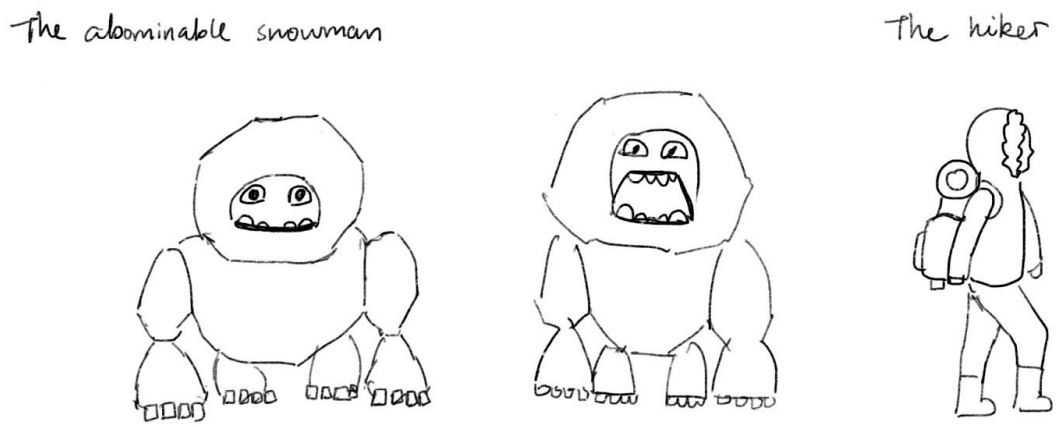
2.3.1 Back Story

You are a mountain-climber who's on a quest to collect all 10 magical gems located in the different ice caves found in the monstrous rock walls of the mountains. However, your ice shoes plummeted down the mountain on your ascent up to the peak so you must use the propulsion from your jet-pack backpack to glide your way through the super slippery cave floors.

2.3.2 Spaces/Worlds



2.3.3 Characters



The abominable snowman is aggressive. The hiker is adventurous and brave.

2.3.4 Levels of Difficulty

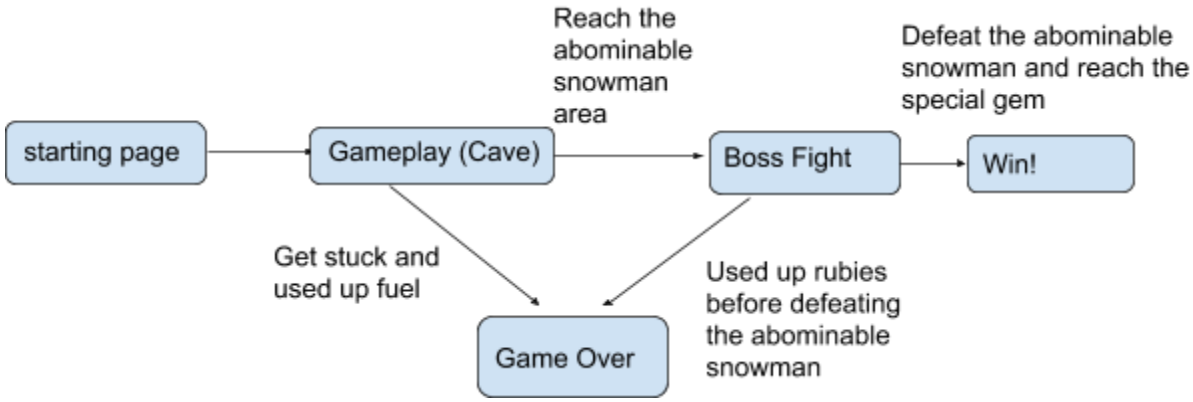
For higher levels, we have more obstacles in the cave and fewer gems.

2.4 Interaction Models

2.4.1 User Interface - Navigation and Movement

The user will control using the mouse and the keypad. Two resource bars are shown at the top of the screen, one tracking diamonds and one tracking rubies.

2.4.2 Game Play Sequence and Levels



2.4.3 User/Environment – Obstacles and Props

The mouse determines the direction in which the player would look. The camera angle is fixed and its distance from the character is constant. So we could calculate where the mouse points to in the world space, and that is the look vector of the character. The length of the mouse click determines how much fuel is used. The fuel gives a force along the direction of the character's look vector. If the character hits an ice pole, he will go off course as the pole gives a force in the collision point's normal direction.

The boss fight will be in first person view, and the player could shoot at the abominable snowman along the camera's look vector. The camera is also controlled by the mouse.

2.4.4 User/Character

The only other character the player will interact with is the abominable snowman. The player can only shoot at it.

2.4.5 Character/Character

The abominable snowman will walk back and forth in front of the crystal. It will roar at you occasionally and move you backwards. You can throw a ruby at it to melt it down.

2.4.6 Puzzle Design

Not featured.

2.4.7 Motion Tracking

Not featured.

2.4.8 Multi-Player

Not featured.

2.4.9 Mobile

Not featured.

2.4.10 Networked Play

Not featured.

2.5 Performance and Scoring

2.5.1 State Variables

What are all the character, environment and gameplay variables necessary to save/restore or pause/resume the game or virtual world experience.

Variables that will be saved:
Location of the player
Rubies number
Diamonds number
Size of abominable snowman

2.5.2 Feedback

Positive Feedback: in the beginning, there will be an abundance of gems. There is also a longer reaction time for the roar from the abominable snowman.
Negative Feedback: the closer you get to the end of the cave, the less gems there are. There is a shorter reaction time for the roar.

2.5.3 Performance and Progress Metrics

You lose by using up all of your gems before collecting the final crystal. You win by collecting the final crystal.

3 GAME DESIGN - IMPLEMENTATION DETAILS

3.1 Design Assumptions

3.1.1 Hardware

The game will run on Mac OS X and Windows 64-bit computers with quad-core Intel or AMD processors, 2.5 GHz or faster, and 8 GB RAM. System should have NVIDIA GeForce 470 GTX or AMD Radeon 6870 HD series graphics card or higher.

3.1.2 Software

We will use Unreal Engine 4 to develop the game. It will be available for Windows 10 and Mac OS X 10.9.2 or later.

3.1.3 Algorithms and Techniques

Algorithms:

- Player velocity: Since the game is heavily inertia-based, the current velocity the player is moving is extremely important. Velocity will be tracked as a Vec3 of floats: when the player hits a certain key, the jetpack will apply a positive acceleration in the direction of the player's forward vector (with upwards component normalized and removed). If the player wishes to fly upward, a different key can be pressed to move the player in the direction of the look vector without the upward component removed. When the player moves forward, the friction of the ice applies a negative force on the player's velocity (the player slows down as they skate on the ice longer), and when flying upward, gravity is pulling the player back down. Crashing into a wall or pillar in the environment will reverse the direction of the inertia (based on physics momentum).
- Gem collection: When a gem is within the reach of the player (the gem position is within a certain radius of the player's position), the gem will glow, indicating the player can collect it. Clicking "F" will collect the fetch the gem and add it to the player's inventory (seen in fuel bar and ruby bar).
- Fuel bar update: Throughout the game, you collect gems to power your jetpack. Whenever you use the jetpack, the fuel bar decreases, and whenever you collect gems, the fuel bar increases. We will track the fuel level using a float variable, increasing and decreasing accordingly with key presses and gem acquisitions. If the fuel level hits 0 and the player's velocity also hits 0 (or almost 0), the player loses.
- Special gem collection bar: In the game, there are special ruby gems that are needed to defeat the boss at the end of the game. There is a status collection bar for this as well showing the percentage of rubies you've acquired so far out of all the rubies available in the level. This is calculated as a float based on an integer

variable representing the number of rubies currently collected and the total number of rubies available.

- Monster roars: There is a roar that echoes through the cave at set intervals which applies a force to the player in the opposite direction of the goal. This roar echoes through the cave and increases in volume as the "echo" gets closer to the player—this gives the player time to prepare for gust of wind from the roar as it approaches. Depending on the player's current position in the cave, we will have a series of different audio recordings (of louder and louder roars) to play as the player gets closer to the source of the roars, and decreasing time intervals set for the player to react to the roar.
- Player obstacles: If a player hides behind an ice pillar (this can be determined by if the player's position is within a certain range of an ice pillar's), the ice pillar blocks the force of the monster's roar from the player and the player does not get pushed backwards. The cave also has cracks / puddles in the ice, and if the player falls into one of these, they will lose 70% of their gems.
- Door busting: If the player hits a door with a large enough velocity, the door will shatter. Otherwise, the player will react as if just hitting a regular obstacle (cavewall, ice pillar).
- Boss battle: To enter the boss battle, the player clicks on the boss in the final cavern, and the third person camera mode switches into first person. The boss battle comprises of a few algorithms.
    - Follow algorithm: The "boss" in the final room will follow the player, blocking them from the final goal. The boss' position will update based on the position of the player.
    - Throwing rubies: The player can hit a key to throw rubies at the ice monster. This will apply a force in the direction of the player's look vector to a ruby.
    - Shrinking: If a ruby hits the boss, he begins to "melt" and decreases in scale by a certain amount. To do this, we will have a model for the monster and simply decrease the model in size every time a ruby hits, or replace this model with a smaller "melting" version of the current model.

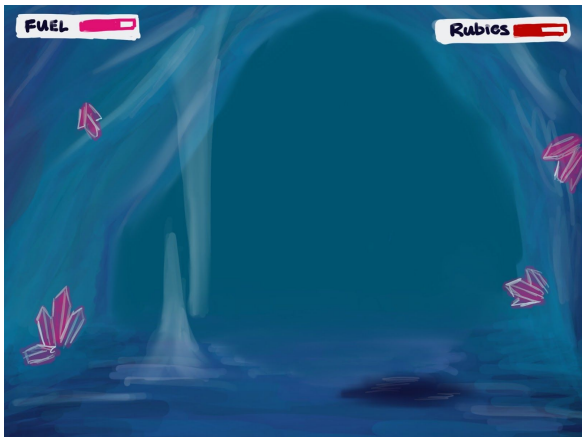
Plug-Ins

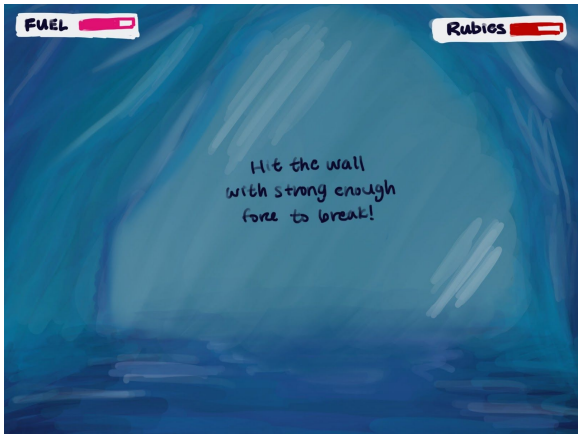
- Unreal has a built-in following algorithm that we will use for the boss
- A following algorithm will also be used for following the player through the cave.

Animations

- Whenever the player smashes through a doorway, there will be a shatter animation applied.

3.2 Storyboards

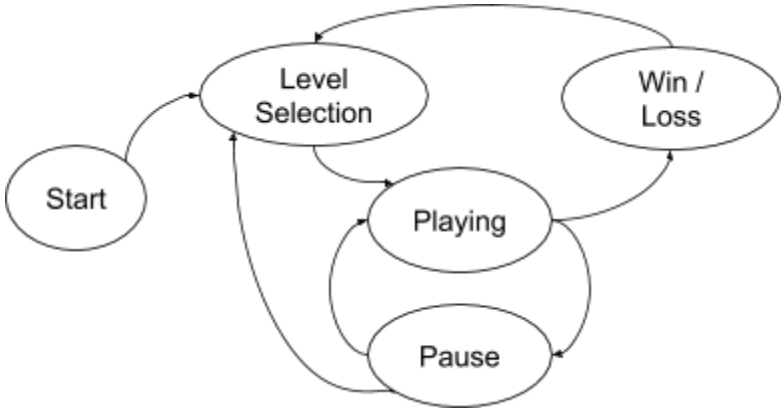




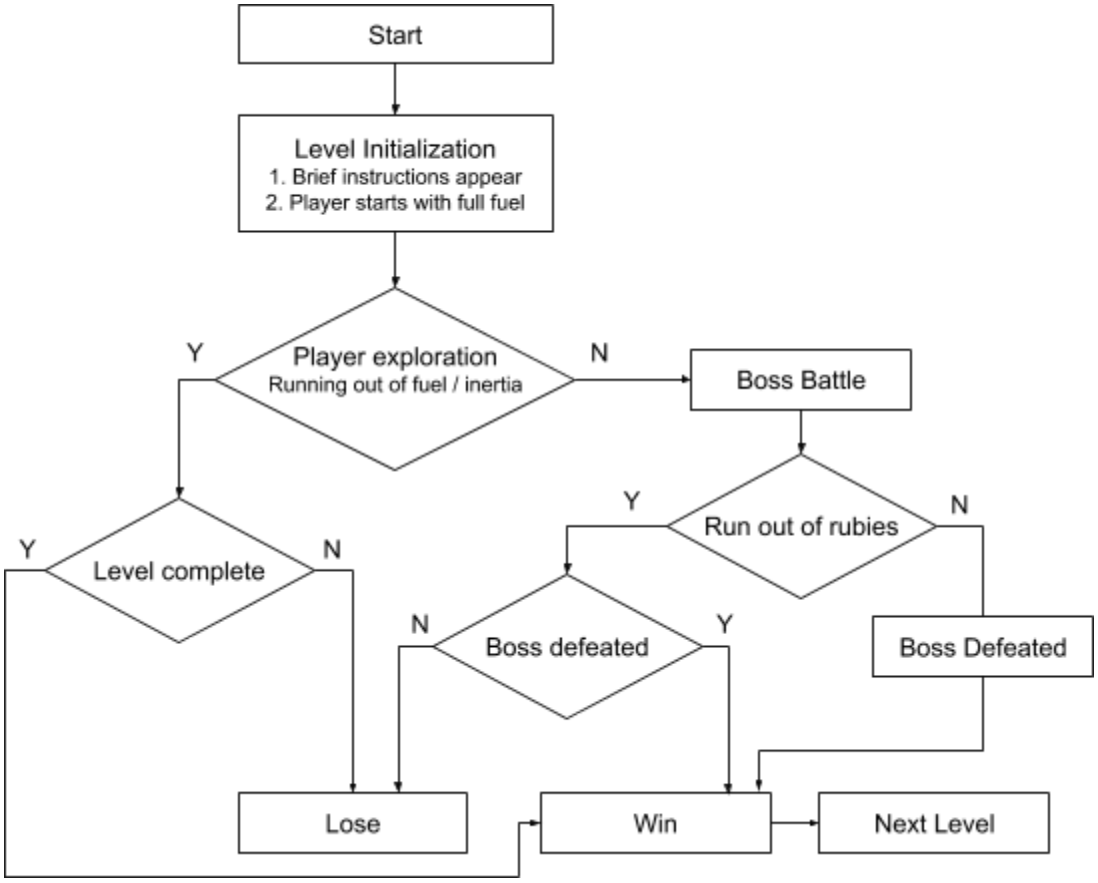
3.3 Design Logic

3.3.1 Finite State Machines – Events/States/Effects

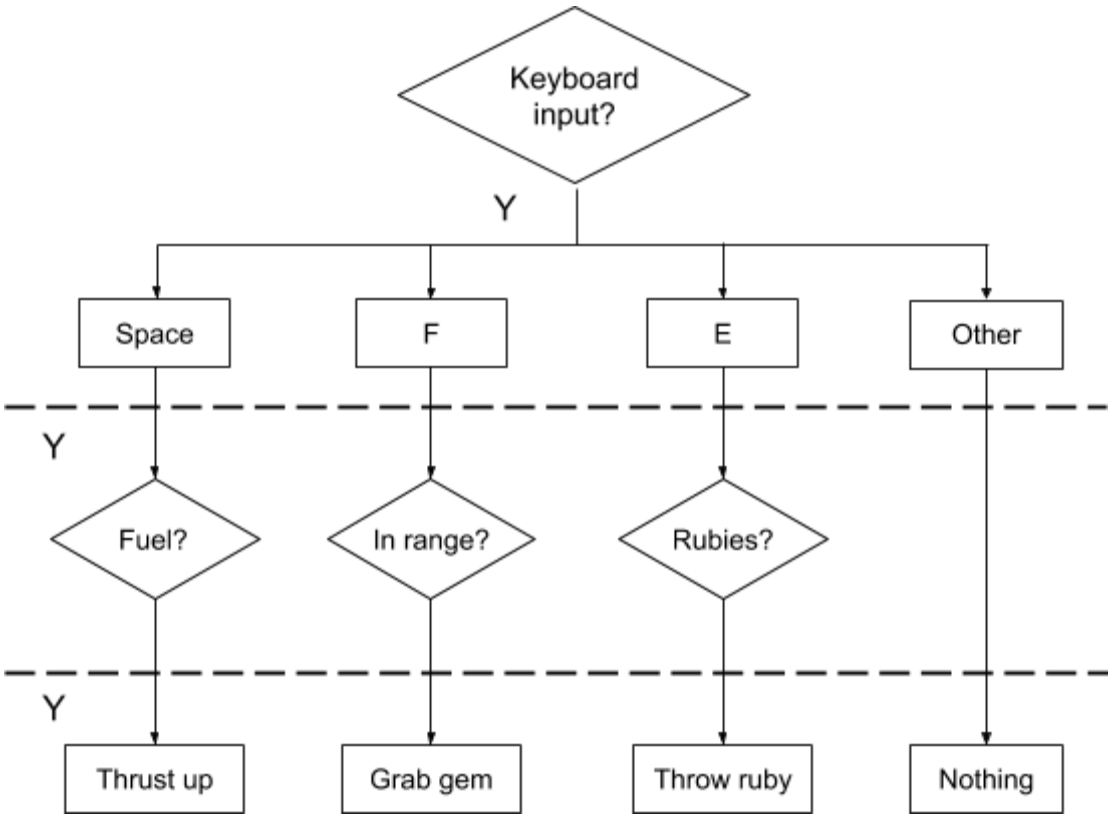
The player starts the game by choosing a level. Once the level is selected, the player enters playing mode. The player can pause at any point and resume, quit, or restart. At any point, if the player runs out of both fuel and inertia or fails to beat the boss, the game is lost. Otherwise, successfully reaching the end and beating the boss results in a win. Later levels can only be reached after earlier levels are beaten.

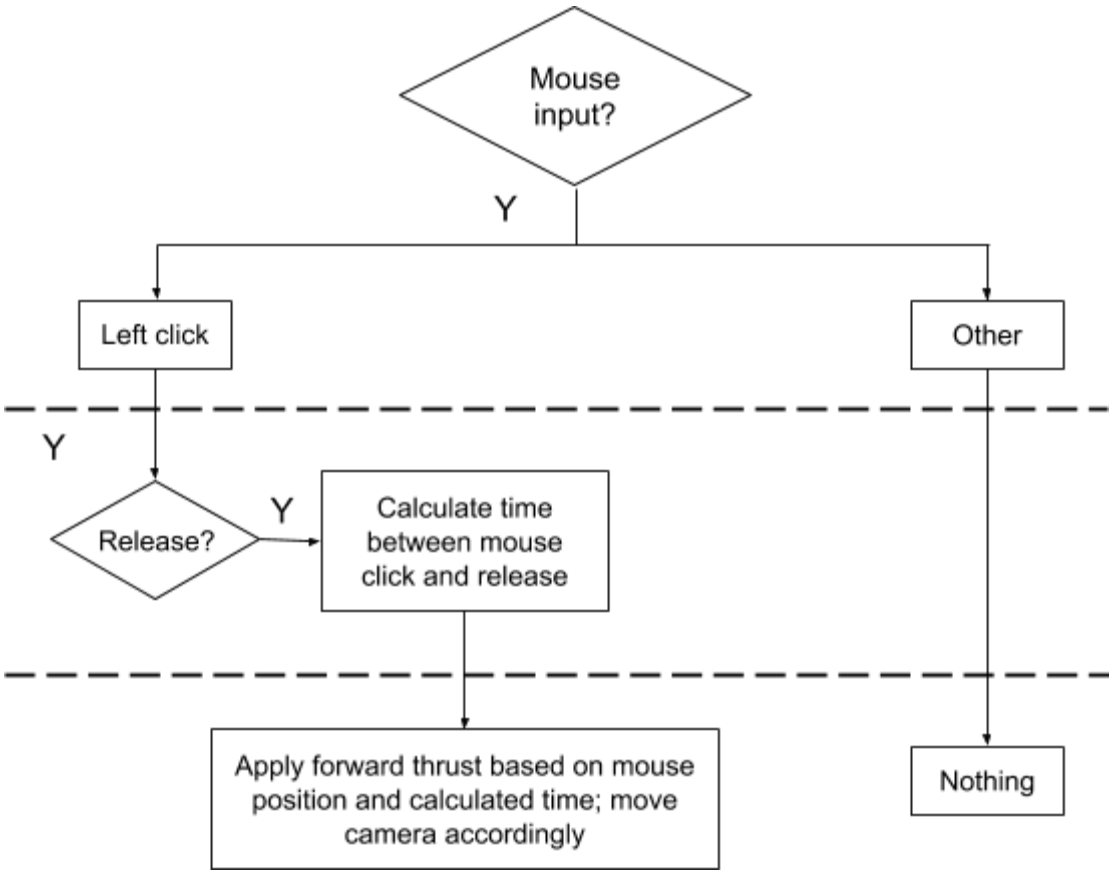


Each level has the following game flow logic:



Players can perform 3 key-triggered actions: propelling upwards, grabbing gems, and throwing gems. Basic moving around is controlled by mouse-clicks: the player clicks where they want the hiker to go, and the length of time they hold down the mouse click for determines how much fuel / thrust is used by the jetpack to move (longer click = higher thrust = faster speed).





3.3.2 User Solution/Actions

The player wins the game by making it to the end of a level and beating the boss without running out of gems or inertia.

3.4 Software Versions

3.4.1 Alpha Version Features (vertical slice through total experience)

The following basic gameplay features will be implemented:

- Player can propel through a simple environment while collecting gems
- Obstacles (pillars, ice ponds) should be present and interact with player correctly
    - Player stops moving when hitting a pillar
    - Player loses percentage of collected gems if entered an ice pond
- Cave walls for players to smash through

3.4.2 Beta Version Features

Some of the more complex mechanics in the game will be implemented, and the major 3d assets will be introduced.

- We will include a fuel monitor, which keeps track of the fuel level of the player
- Introduce upward propulsion

- Import 3D gem models
    - Differentiate between different types of gems: add rubies
- Snowman boss battle

3.4.3 Descriptions of any tutorial levels and/or self-running Demos

None

4 WORK PLAN

4.1 Tasks

		Description	Time	Member
1	**Task: Cave Navigation**	**All necessary movements in the cave.**	**7 days**	AC
1.1	Subtask: Floor Navigation	The player should be able to glide on the cave floor.	4 day	AC
1.2	Subtask: Wall Navigation	The player should be able to fly up the cave walls.	3 day	AC
2	**Task: Gem Collection**	**Player should be able to collect the gems located in the cave.**	**3 day**	AC/EL
2.1	Subtask: Collision Detection	If a player is within a certain radius of the gem, they can collect the gem, also the gem changes color	3 day	AC/EL
3	**Task: UI**	**Making the UI be user friendly and look nice.**	**4 days**	SX
3.1	Subtask: Make the in-game screen/UI	Make the level user friendly.	2 days	SX
3.2	Subtask: Making the end game screen	Make both screens where the player wins and loses	2 days	SX
4	**Task: Collisions**	**Make sure the appropriate actions result from certain collisions**	**5 days**	EL

4.1	Subtask: Cave Wall/Pillars	Make sure the player cannot walk through cave walls/pillars and can destroy the cave walls	1 day	EL
4.2	Subtask: Frozen Ponds	Make sure the game recognizes collisions with frozen ponds and loses gems when occurs	2 day	EL
4.3	Subtask: Snowman Roars	Make sure if that the snowman roar pushes the player back in the cave unless located behind a ice pillar/rock/wall	2 days	EL
5	**Task: Snowman Battle**	**Make the snowman smaller when a gem is thrown at it**	**3 day**	SX
6	**Task Level Design**	**Design the gameplay level**	**3 days**	AC, SX, EL
6.1	Subtask: Tuning	Tune how many resources the player should get at each interval of game play	3 days	AC, SX, EL
7	**Task: Effects**	**All Visual and Audio Effects**	**16 days**	AC, SX, EL
7.1	Subtask: Modeling, Shading, Lighting and Animation	Make the cave model, create an ice shader, light the cave, add snow particles in the air, and animate the abominable snowman	14 days	AC, SX, EL
7.2	Subtask: Audio	Make sound effects for different collisions and background music	2 days	AC, SX, EL
8	**Task: Scoring/GameTesting**	**Score the player and make sure the components work**	**5 days**	AC, SX, EL

4.2 Milestones

4.2.1 Minor

- Cave Navigation
- Gem Collection
- Game UI
- Cave/Environment Collisions
- Abominable Snowman Battle

- Effects
- Modeling
- Animation
- Shading
- Lighting
- Sound Effects

4.2.2 Major

Alpha Version

The alpha version should just have the core mechanics of the game - namely, the player should be able to propel themselves through a simple environment while collecting gems. The player should lose a percentage of their collected gems if they enter an ice pond, should not be able to walk through the cave walls or pillars, and should be able to “melt” the abominable monster and collect the goal-driven gem at the end of the level.

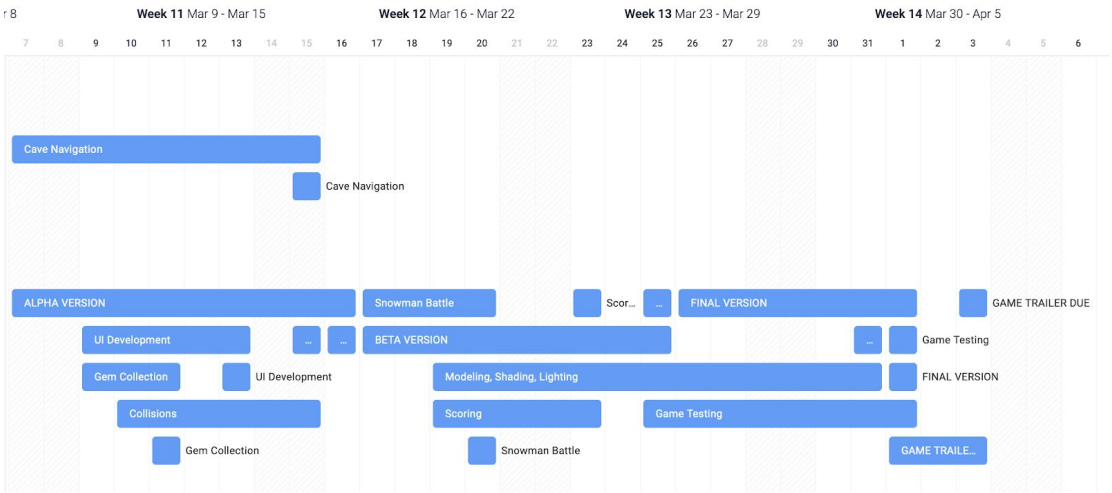
Beta Versions

The beta version should have more complex mechanics in the game. Through playtesting we will include a fuel monitor which will calculate the fuel used and gained from the gems in the cave. There will be the introduction of major 3d assets to transport the player into the ice cave. There will be some basic animation and audio.

4.3 Development Schedule

Ice Gems Gantt Chart

Things to do	Owner	Due date	Timeline
Cave Navigation	AC	Mar 15	Mar 7 - 15
Collisions		Mar 15	Mar 10 - 15
UI Development		Mar 13	Mar 9 - 13
Gem Collection		Mar 11	Mar 9 - 11
Snowman Battle		Mar 20	Mar 17 - 20
ALPHA VERSION		Mar 16	Mar 7 - 16
Modeling, Shading, Lighting		Mar 31	Mar 19 - 31
Scoring		Mar 23	Mar 19 - 23
BETA VERSION		Mar 25	Mar 17 - 25
Game Testing		Apr 1	Mar 25 - Apr 1
FINAL VERSION		Apr 1	Mar 26 - Apr 1
GAME TRAILER DUE		Apr 3	Apr 1 - 3
+ Add			
			Mar 7 - Apr 3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Week 11 Mar 9 - Mar 15
Week 12 Mar 16 - Mar 22
Week 13 Mar 23 - Mar 29
Week 14 Mar 30 - Apr 5
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Cave Navigation
Cave Navigation
ALPHA VERSION
Snowman Battle
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